

Exam 1 Review Worksheet

Problem Solving, Math 1000 – Summer 2016

Test Friday, May 27th

Problem 1 Fill in the blanks with the definition of the term on the left:

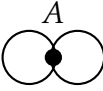
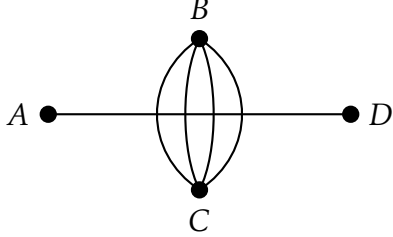
<i>Graph</i>	<i>A graph is a set of vertices and a set of edges</i>
<i>Edge Set</i>	<i>The edge set is the set of all the edges in the graph</i>
<i>Vertex Set</i>	<i>The vertex set is the set of all the vertices in the graph</i>
<i>Adjacent</i>	<i>Two vertices in a graph are adjacent if they are connected by an edge</i>
<i>Loop</i>	<i>A loop is an edge from a vertex to itself</i>
<i>Degree</i>	<i>The degree of a vertex is the number of edges that meet at that vertex</i>
<i>Odd Vertex</i>	<i>An odd vertex is a vertex with odd degree</i>
<i>Even Vertex</i>	<i>An even vertex is a vertex with even degree</i>
<i>Multiple Edges</i>	<i>A graph has multiple edges if there exists more than one edge between a pair of vertices</i>
<i>Simple Graph</i>	<i>A simple graph is a graph without loops or multiple edges</i>

<i>Complete Graph</i>	<i>A complete graph is a simple graph with all possible edges</i>
<i>Circuit</i>	<i>A circuit is a sequence of edges, each adjacent to the next, with no edge included more than once, which starts and ends at the same vertex</i>
<i>Path</i>	<i>A path is a sequence of edges, each adjacent to the next, with no edge included more than once, which starts and ends at different vertices</i>
<i>Disconnected Graph</i>	<i>A disconnected graph is a graph where there is not at least one path between any two given vertices. That is to say, a disconnected graph is a graph with more than one component</i>
<i>Euler Circuit</i>	<i>An Euler circuit is a circuit that includes every edge in the graph</i>
<i>Euler Path</i>	<i>An Euler path is a path which includes every edge in the graph</i>
<i>Hamilton Circuit</i>	<i>A Hamilton circuit is a circuit which visits each vertex in the graph once and only once</i>
<i>Hamilton Path</i>	<i>A Hamilton path is a path which visits each vertex in the graph once and only once</i>

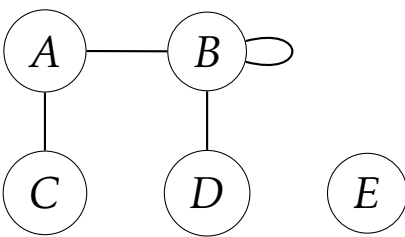
Problem 2 What is the difference between a bridge in a graph and a bridge over water?

A bridge in a graph is an edge in a connected graph that, if removed, would disconnect the graph. A bridge over water is a large and generally durable structure which allows people or vehicles to travel safely across an expanse of liquid.

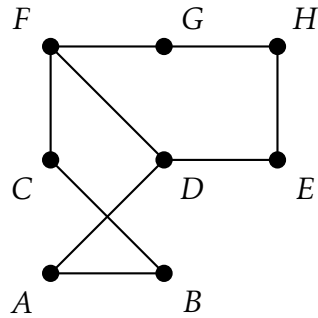
Problem 3 Fill in the blanks based on the given information.

Graph:		
Give the Edge Set:	$E(G) = \{AA, AA\}$	$E(G) = \{AD, BC^4\}$
Give the Vertex Set:	$V(G) = \{A\}$	$V(G) = \{A, B, C, D\}$
Is the graph connected?	yes	no
List of all bridges:	none	AD
List the degree of each vertex:	$\deg(A) = 4$	$\deg(A)=1, \deg(D)=1, \deg(C)=4, \deg(B)=4$
Sum of degrees of the graph:	4	10
Exact number of odd vertices:	Even: 1 Odd: 0	Even: 2 Odd: 2
Is there an Euler Circuit (EC), an Euler Path (EP), or neither?	Euler Circuit	Euler Path

Problem 4 Fill in the blanks. If there is a blank for the graph, draw something that is reasonable, based on the rest of the information.

(One possible) Graph:		This graph cannot exist! We can't have an odd number of odd vertices by Euler's sum of degrees theorem.
Give the Edge Set:	$E(G) = \{AB, AC, BB, BD\}$	
Give the Vertex <u>Set</u> :	$V(G) = \{A, B, C, D\}$	
Is the graph connected?	no	
List of all bridges:	none	AB
List the degree of each vertex:	$\deg(A) = 2, \deg(B) = 4,$ $\deg(C) = 1, \deg(D) = 1,$ $\deg(E) = 0$	$\deg(A) = 1$
Sum of degrees:	8	12
Exact number of odd vertices:	2	3
Is there an Euler Circuit (EC), an Euler Path (EP), or neither?	Euler Path	

Problem 5 Given the graph below, find a Hamilton/Euler Circuit/Path (that is, list the vertices in the order they would be traveled). If there isn't one, explain why not. If there is more than one, just list one, but say how many there are.



EC:

None, there is more than one vertex of odd degree.

EP:

Six. The following three and their mirror images:

1. F, G, H, E, D, F, C, B, A, D
2. F, G, H, E, D, A, B, C, F, D
3. F, C, B, A, D, E, H, G, F, D

HC:

F, G, H, E, D, A, B, C, F or any shifting thereof, for a total of eight.

HP:

F, G, H, E, D, A, B, C, or any similar removal of the final edge, for a total of seven.

What (if anything) changes if another edge FD is added?

Nothing for Hamilton paths and circuits – we care about visiting vertices more than once, so an extra edge where we already have one won't help. However, we do then lose our two odd vertices, giving us an Euler circuit and no Euler paths.

Problem 6 Draw a graph in each box so that it is possible to find the Path/Circuit listed (first), but not possible to find the Path/Circuit prohibited (second), as specified. That is, the first graph must have at least one EC, but no HC is possible.

EC, no HC	EP, no HP	HC, no EC	HP, no EP	HP, no HC

There are many, many possible solutions to these problems. Refer to the table we drew in class for solutions – the answers should follow readily from there.

Problem 7 What is the degree of each vertex in a K_n ? How many edges does a K_n have? How did you find this formula?

The degree of each vertex in a K_n is $n-1$. A K_n has $\frac{n(n-1)}{2}$ edges. This formula is derived by taking the sum of the degrees and dividing by two, as given by Euler's sum of degrees theorem.

Problem 8 How many Hamilton paths does a K_n have? How many Hamilton circuits? How did you arrive at this answer?

A K_n has $n!$ Hamilton paths and $(n-1)!$ Hamilton circuits.

Problem 9 How many Euler paths, Euler circuits, Hamilton paths, and Hamilton circuits does the graph K_6 have?

A K_6 has 6 odd vertices, so it has neither Euler paths nor Euler circuits. A K_6 has $6!$ Hamilton paths and $(6-1)!$ Hamilton circuits.